Retail Sales Performance Analysis

BI Capstone Final Presentation

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Introduction

Introduction

Business scenario

This project analyzes retail sales data for a chain of stores. The aim is to understand sales performance across products, locations, and time.

Purpose of the analysis

The goal was to turn raw data into useful business insight through cleaning, structured analysis, visualization, and summary reporting.

What this presentation covers

This presentation explains the process followed across the capstone and summarizes the main findings, insights, and recommendations.

Methodology

Data Details

Data used in the project

- sales.csv: transaction-level sales, revenue, stock, price, and promotion fields
- product_hierarchy.csv: product grouping and hierarchy structure
- store_cities.csv: store and city mapping
- store_names.csv: store name lookup
- city_names.csv: city name lookup
- product_names.csv: product name lookup

Why this matters

Together, these files provide transaction data, product structure, store information, and location context.

This combination made it possible to analyze performance by product, by location, and over time.

Module 1: Data Cleaning and Preparation Using Excel

Objective

Prepare the raw data for reliable analysis.

What was done

- Imported the source files into one workbook
- Checked row and column structure
- Reviewed missing values
- Standardized dates, numbers, and text
- Removed duplicate records
- Prepared the data for analysis

Output

A clean and consistent dataset ready for pivot tables and later analysis.

Module 2: Data Querying and Analysis Using PostgreSQL

Objective

Move the prepared data into a structured database environment for deeper analysis.

What was done

- Loaded cleaned data into PostgreSQL
- Validated that the tables were populated correctly
- Queried the data using SQL
- Summarized performance across categories, locations, and time
- Used hierarchical summaries and trend analysis

Output

A more structured and repeatable analysis process that supported deeper insight.

Module 3: Data Visualization and Statistical Analysis

Objective

Use visual and statistical methods to better understand patterns in the data.

What was done

- Created charts to compare trends, categories, and relationships
- Used different visual formats to show the data from multiple angles
- Ran a simple regression to test whether date meaningfully predicts sales

Output

A clearer view of patterns in the data and a basic statistical check of time-based relationship strength.

Module 4: Data Visualization and Dashboards Using Tableau

Objective

Communicate findings more clearly through interactive visual analysis.

What was done

- Built core visualizations in Tableau
- Compared performance across categories and time
- Combined views into a dashboard
- Added filter-based interactivity

Output

A more accessible presentation of insights that supports focused decision-making.

Results

Module 1, Lesson 1: Data Cleaning and Preparation

product_id	store_id	date	sales	revenue	stock	price	promo_type_1	promo_bin_1	promo_type_2
P0001	S0002	2017-02-01	0	0.00	8	6.25	PR14		PR03
P0001	S0002	2017-03-01	0	0.00	8	6.25	PR14		PR03
P0001	S0002	2017-04-01	0	0.00	8	6.25	PR14		PR03
P0001	S0002	2017-05-01	0	0.00	8	6.25	PR14		PR03
P0001	S0012	2017-02-01	1	5.30	0	6.25	PR14		PR03
P0001	S0013	2017-02-01	2	10.59	0	6.25	PR14		PR03
P0001	S0023	2017-02-01	0	0.00	6	6.25	PR14		PR03
P0001	S0023	2017-03-01	0	0.00	6	6.25	PR14		PR03
P0001	S0023	2017-04-01	0	0.00	6	6.25	PR14		PR03
P0001	S0023	2017-05-01	0	0.00	6	6.25	PR14		PR03
P0001	S0025	2017-02-01	0	0.00	1	6.25	PR14		PR03
P0001	S0025	2017-03-01	0	0.00	1	6.25	PR14		PR03
P0001	S0025	2017-04-01	0	0.00	1	6.25	PR14		PR03
P0001	S0025	2017-05-01	0	0.00	1	6.25	PR14		PR03
P0001	S0027	2017-02-01	0	0.00	7	6.25	PR14		PR03
P0001	S0027	2017-03-01	0	0.00	7	6.25	PR14		PR03
P0001	S0027	2017-04-01	0	0.00	7	6.25	PR14		PR03
P0001	S0027	2017-05-01	1	5.30	6	6.25	PR14		PR03
P0001	S0040	2017-02-01	0	0.00	19	6.25	PR14		PR03
P0001	S0040	2017-03-01	0	0.00	19	6.25	PR14		PR03
P0001	S0040	2017-04-01	0	0.00	19	6.25	PR14		PR03
P0001	S0040	2017-05-01	0	0.00	19	6.25	PR14		PR03
P0001	S0049	2017-02-01	0	0.00	8	6.25	PR14		PR03
P0001	S0049	2017-03-01	0	0.00	8	6.25	PR14		PR03

sales | product_hierarchy | product_names | store_cities | store_names

Ready | Accessibility: Investigate | Count: 49186 | 100%

What this shows

The raw data was cleaned by checking structure, handling missing values, standardizing dates and numeric formats , trimming extra space from text fields and removing duplicate records.

Why it matters

This improved consistency and reduced the risk of errors in the later analysis.

Module 1, Lesson 2: Data Analysis Using Pivot Tables

Sales overview

Store Name	Total Sales	Total Stock	Total Revenue	Average Sales
DIGI	5538	133536	$ 18,146.50	0.68
Electro World (National Chain)	3675	153521	$ 17,880.98	0.38
Darty	1901	45375	$ 5,080.29	0.64
Elettrodomestici Rossi	1247	71470	$ 4,545.47	0.25
Currys (National Chain)	389	31473	$ 1,055.21	0.15
Euronics Lisboa (National Chain)	398	30929	$ 658.68	0.25
Grand Total	**13148**	**466304**	**$ 47,367.13**	**0.44**

Sales by city

City Name	Total Sales	Total Revenue	Average Sales
Edinburgh	5538	$18,146.50	0.68
Helsinki	3675	$17,880.98	0.38
London	1901	$5,080.29	0.64
Saint Petersburg	1247	$4,545.47	0.25
Vienna	787	$1,713.89	0.19
Grand Total	**13148**	**$47,367.13**	**0.44**

Product performance

Product Name	Total Revenue	Total Sales	Average Sales
Portable Heater	$14,378.98	637	3.28
Handheld Milk Frother	$2,988.52	1218	6.09
SolarGrillpulse	$1,123.66	276	1.15
PowerDryerflow	$966.95	867	6.83
SolarBlenderlux	$846.66	273	2.03
NeoHeatergen	$806.80	101	0.51
SmartFridgewave	$800.06	338	1.40
CyberTVwave	$790.66	40	0.20
MaxDryerwave	$736.60	150	0.75
TechVacuumflow	$698.85	93	0.54
TechFridgematic	$511.26	184	0.93
TechMixerflow	$459.99	44	0.29
Hardwood Floor Cleaners	$455.31	27	0.30
Smart Speakers (with voice assistants)	$440.13	700	2.83
NanoFridgeflow	$433.39	36	0.18
EcoVacuumcast	$417.94	244	1.01
Immersion Blender	$397.54	151	1.03
Grand Total	**$47,367.13**	**13148**	**0.44**

What this shows

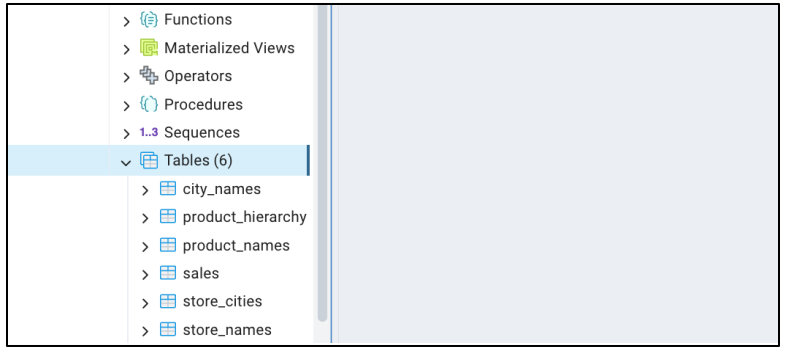
These summaries helped compare overall sales, location performance, and stronger versus weaker Products.

Why it matters

This supports decisions on product focus, city-level planning, and where the business should investigate weaker performance.

Module 2, Lesson 1: Data Querying Using PostgreSQL

PgAdmin - tables



SQL table populating query

Query Query History

```
SELECT * FROM sales LIMIT 5;
```

Data Output Messages Notifications

	product_id character varying	store_id character varying	date date	sales double precision	revenue double precision
1	P0001	S0002	2017-02-01	0	0
2	P0001	S0012	2017-02-01	1	5.3
3	P0001	S0013	2017-02-01	2	10.59
4	P0001	S0023	2017-02-01	0	0
5	P0001	S0025	2017-02-01	0	0

SQL Sales Performance query

Query Query History

```
SELECT
  p.product_id,
  c.store_id,
  c.city_id,
  SUM(sl.sales) AS total_sales,
  SUM(sl.revenue) AS total_revenue,
  AVG(sl.price) AS average_price
FROM sales sl
INNER JOIN product_hierarchy p
  ON sl.product_id = p.product_id
INNER JOIN store_cities c
  ON sl.store_id = c.store_id
GROUP BY
  p.product_id,
  c.store_id,
  c.city_id;
```

Data Output Messages Notifications

	product_id character varying	store_id character varying	city_id character varying	total_sales double precision	total_revenue double precision	average_price double precision
1	P0001	S0002	C007	0	0	6.25
2	P0001	S0012	C005	1	5.3	6.25
3	P0001	S0013	C026	2	10.59	6.25
4	P0001	S0023	C008	0	0	6.25
5	P0001	S0025	C024	0	0	6.25
6	P0001	S0027	C022	0	0	6.25
7	P0001	S0040	C017	0	0	6.25
8	P0001	S0049	C031	0	0	6.25
9	P0001	S0050	C014	0	0	6.25
10	P0001	S0051	C027	0	0	6.25
11	P0001	S0055	C014	0	0	6.25

What this shows

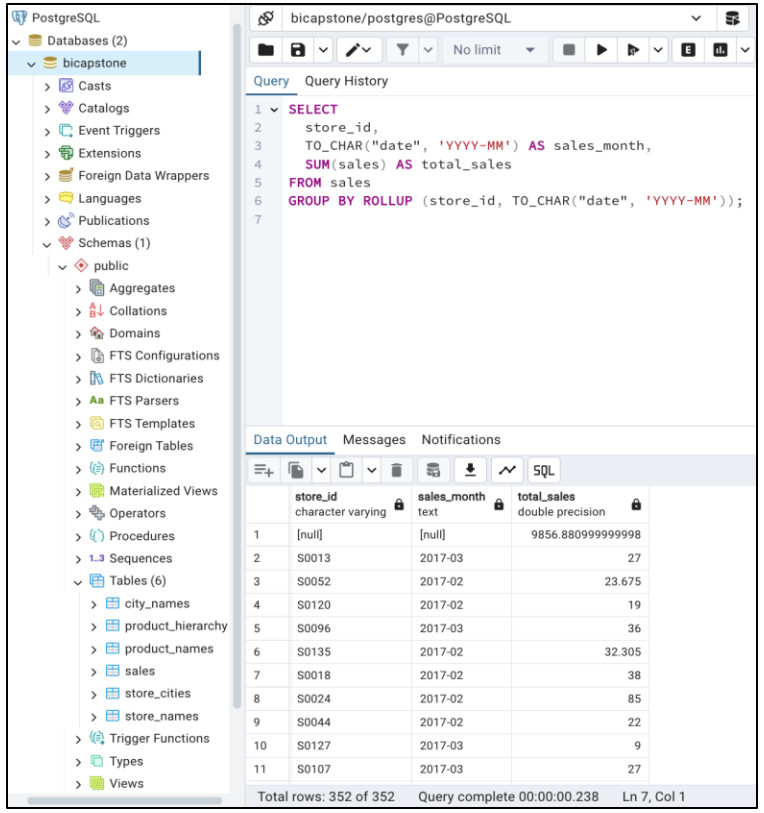
The data was successfully loaded into a structured database and queried for validation and performance analysis.

Why it matters

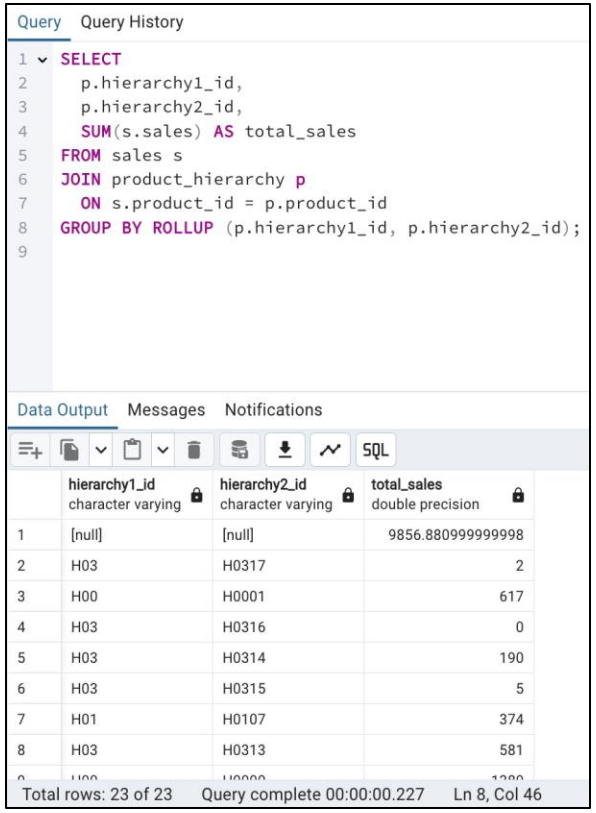
This created a more reliable and repeatable basis for deeper analysis than a spreadsheet-only approach.

Module 2, Lesson 2: Data Analysis Using PostgreSQL

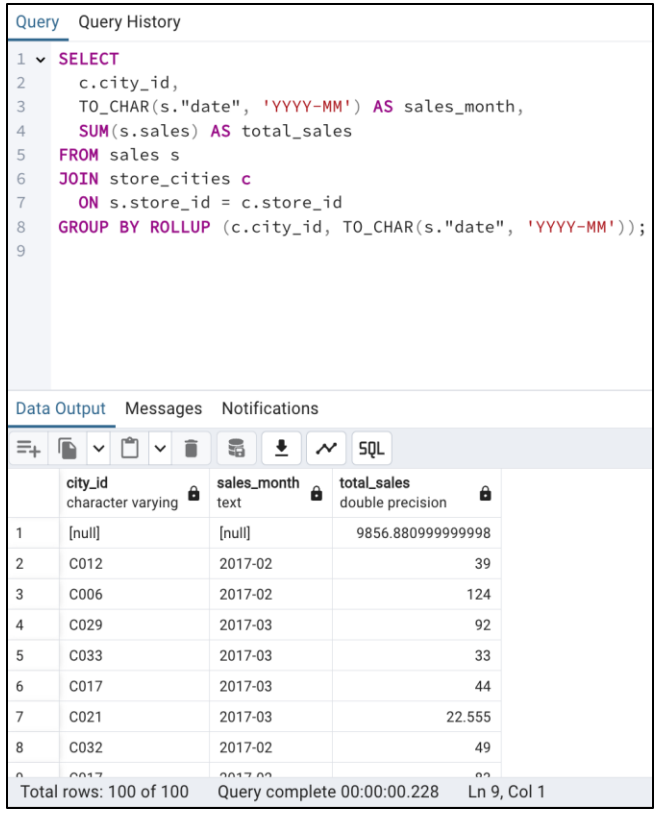
ROLLUP / cube results



Hierarchy Summary



Sales trends over time / regions



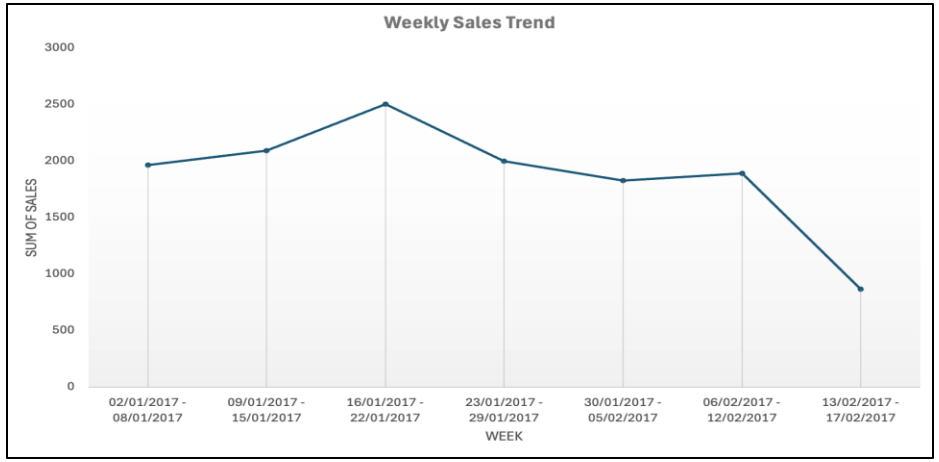
What this shows

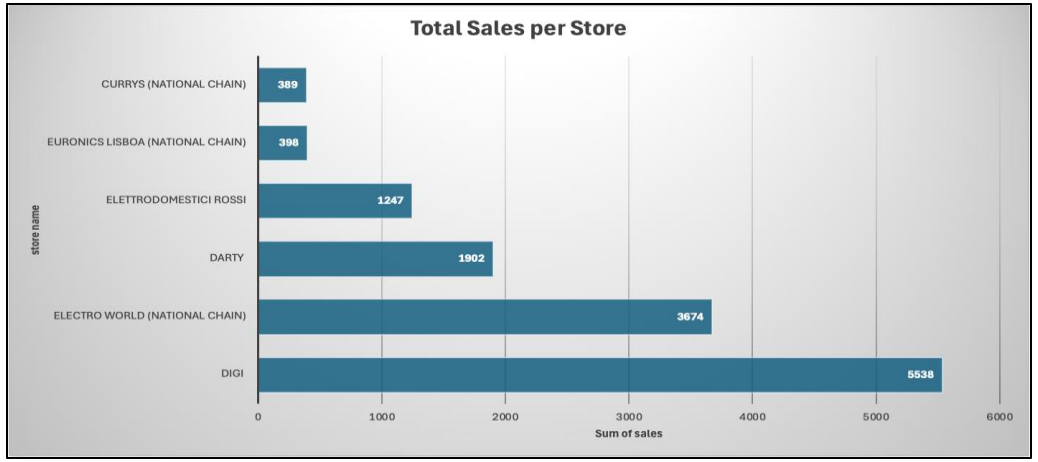
SQL was used to summarize the data across levels, hierarchies, and time periods.

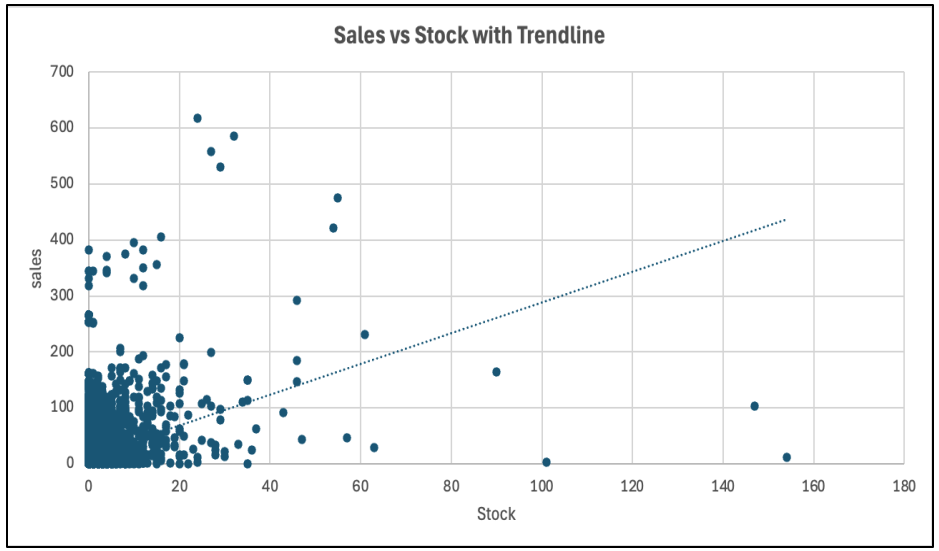
Why it matters

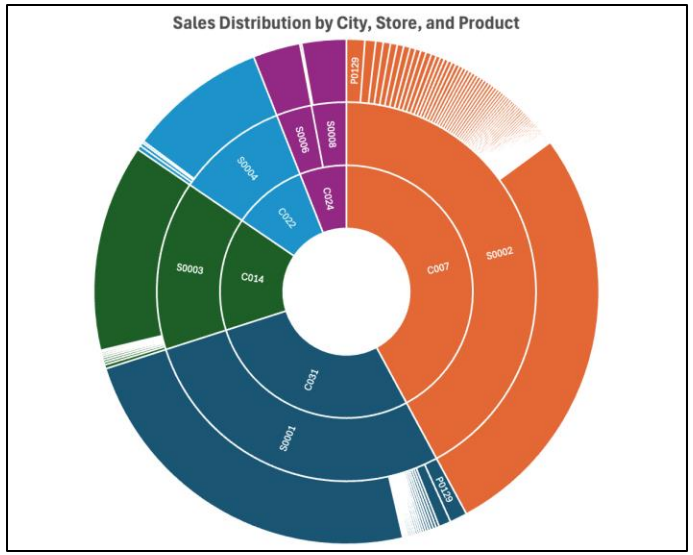
These summaries help the business understand demand patterns, compare regions, and support better stock and sales planning.

Module 3, Lesson 1: Data Visualization Using Excel









What this shows

The charts present the data from multiple angles, including change over time, category differences, hierarchy structure, and variable relationships.

Why it matters

Visual analysis makes patterns easier to understand and explain than raw tables alone.

Module 3, Lesson 2: Statistical Analysis

Regression Analysis - Output

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.00255527							
R Square	6.52941E-06							
Adjusted R Square	-4.34753E-05							
Standard Error	2.805644826							
Observations	20000							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	1	1.027846918	1.027846918	0.130575908	0.717839481			
Residual	19998	157417.1145	7.87164289					
Total	19999	157418.1424						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95%	Upper 95%
Intercept	-23.25708309	65.83469922	-0.353264819	0.723893736	-152.298529	105.7843628	-152.298529	105.7843628
date_serial	0.000556371	0.001539689	0.361352883	0.717839566	-0.002461547	0.003574289	-0.002461547	0.003574289

Interpretation

The p-value is high, which suggests there is no statistically reliable relationship between date and sales in this sample.

The R-squared value is extremely low, which means date explains almost none of the variation in sales.

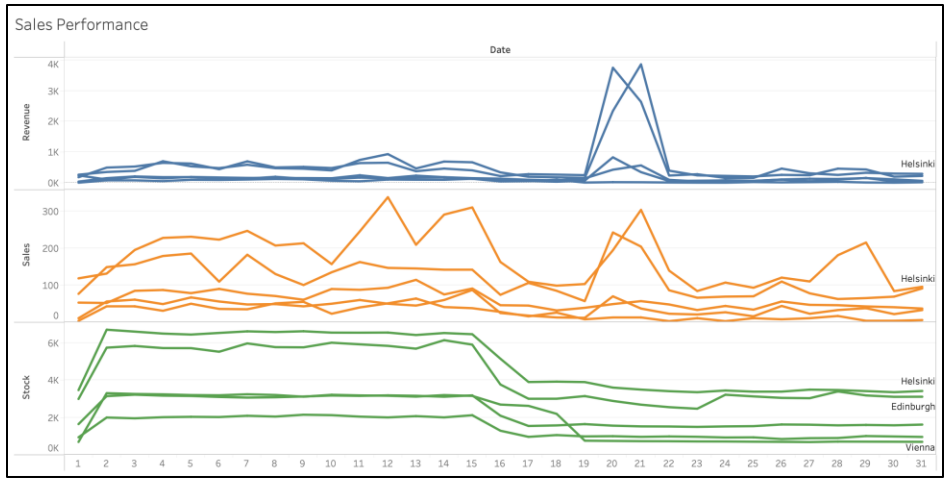
Implication

Date alone does not appear to be a useful predictor of sales here.

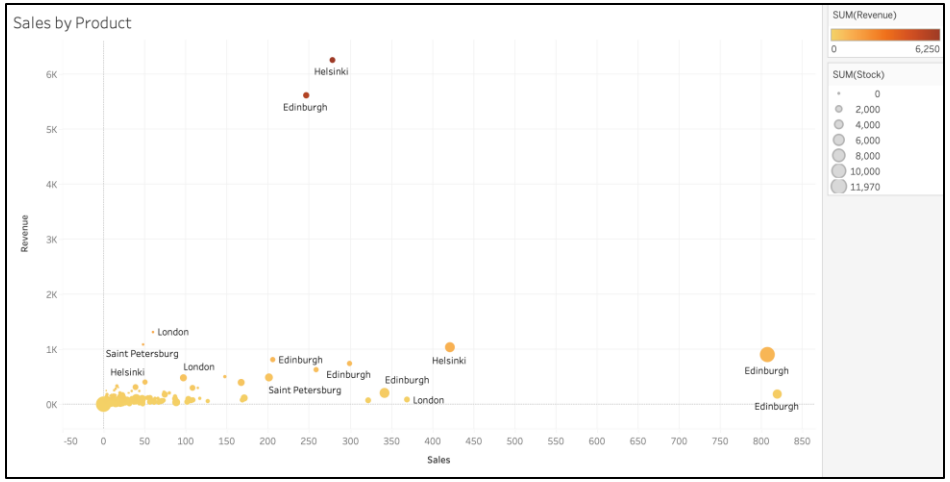
It is likely that sales are influenced more by other factors.

Module 4, Lesson 1: Basic Tableau Visualizations

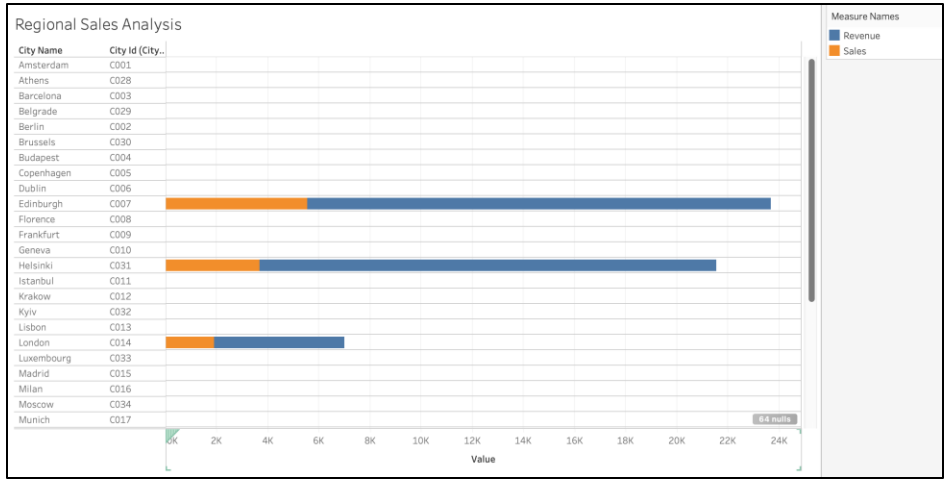
Trend



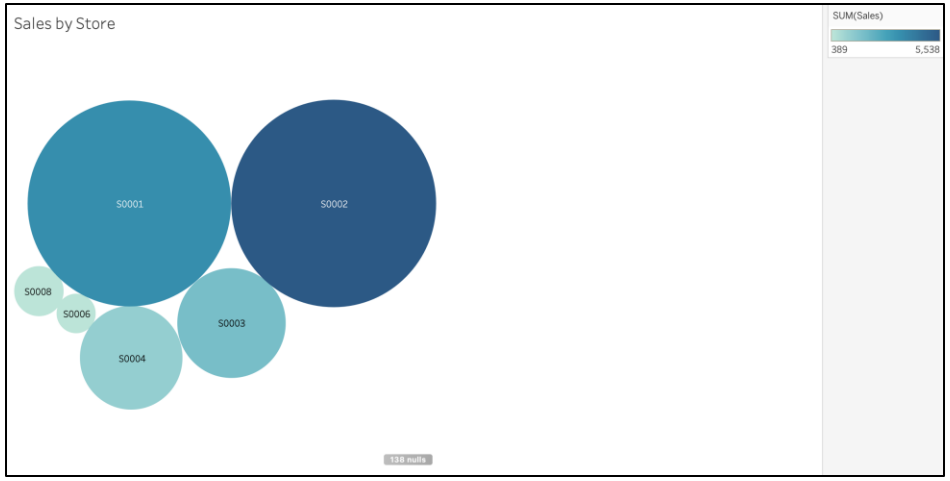
Relationship



Comparison



Relative size



What this shows

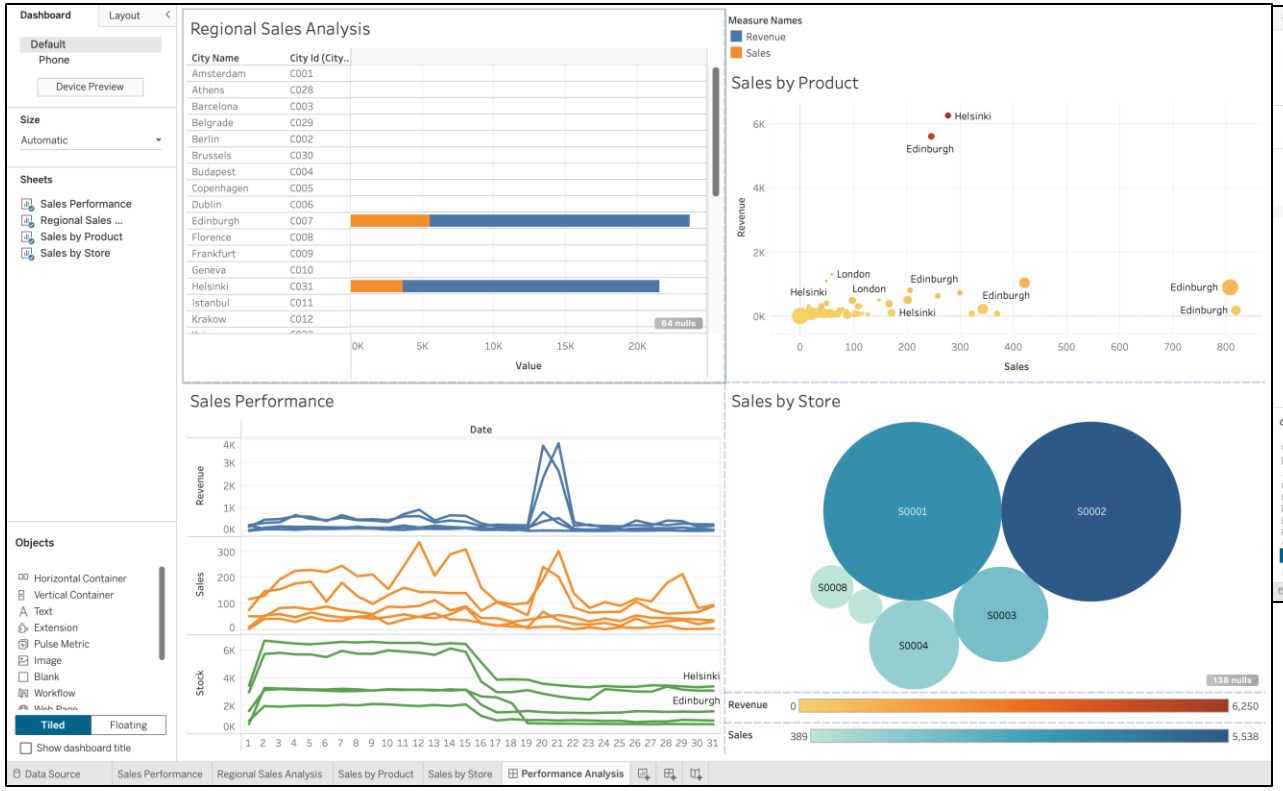
These views make it easier to compare groups, follow patterns over time, and understand relationships and relative contribution.

Why it matters

They support clearer communication of insight to stakeholders.

Module 4, Lesson 2: Advanced Visualizations Using Tableau

Dashboard



Dashboard - filtered

What this shows

The dashboard combines multiple views in one place and allows the user to focus on selected segments through filtering.

Why it matters

This supports faster, more focused decision-making.

Discussion

Insights and Recommendations

Key Insights

- Sales performance was not evenly distributed across locations.
- Product performance was concentrated in a smaller set of stronger items or categories.
- Visual analysis made performance differences easier to identify and explain.
- The regression result suggests that time alone was not a strong predictor of sales.

Recommendations

- Focus planning on the products and locations that contribute most strongly to results.
- Review weaker areas for possible pricing, assortment, or local execution issues.
- Use dashboard-based review to support regular performance tracking.
- Extend future analysis with more drivers, such as promotions, seasonality, or store type.

Conclusion

Summary

- This project transformed raw retail data into a structured and usable analysis.
- The work covered data cleaning, summary analysis, SQL-based querying, chart creation, regression analysis, and dashboard design.
- Together, these steps helped reveal patterns in sales, product performance, and location-based results, and supported practical recommendations for better retail decision-making.

Appendix

Appendix 1

Data before cleaning – 50000 records

product_id	store_id	date	sales	revenue	stock	price	promo_type_1	promo_bin_1	promo_type_2	promo_bin_2	promo_discount_2	promo_discount_type_2
P0001	S0002	2017-02-01	0	0.00	8	6.25	PR14		PR03			
P0001	S0012	2017-02-01	1	5.30	0	6.25	PR14		PR03			
P0001	S0013	2017-02-01	2	10.59	0	6.25	PR14		PR03			
P0001	S0023	2017-02-01	0	0.00	6	6.25	PR14		PR03			
P0001	S0025	2017-02-01	0	0.00	1	6.25	PR14		PR03			
P0001	S0027	2017-02-01	0	0.00	7	6.25	PR14		PR03			
P0001	S0040	2017-02-01	0	0.00	19	6.25	PR14		PR03			
P0001	S0049	2017-02-01	0	0.00	8	6.25	PR14		PR03			
P0001	S0050	2017-02-01	0	0.00	5	6.25	PR14		PR03			
P0001	S0051	2017-02-01	0	0.00	6	6.25	PR14		PR03			
P0001	S0055	2017-02-01	0	0.00	6	6.25	PR14		PR03			
P0001	S0056	2017-02-01	1	5.30	6	6.25	PR14		PR03			
P0001	S0062	2017-02-01	0	0.00	2	6.25	PR14		PR03			
P0001	S0063	2017-02-01	0	0.00	7	6.25	PR14		PR03			
P0001	S0066	2017-02-01	0	0.00	1	6.25	PR14		PR03			
P0001	S0078	2017-02-01	0	0.00	6	6.25	PR14		PR03			
P0001	S0082	2017-02-01	0	0.00	4	6.25	PR14		PR03			
P0001	S0083	2017-02-01	0	0.00	10	6.25	PR14		PR03			
P0001	S0087	2017-02-01	0	0.00	10	6.25	PR14		PR03			
P0001	S0094	2017-02-01	0	0.00	6	6.25	PR14		PR03			
P0001	S0096	2017-02-01	0	0.00	2	6.25	PR14		PR03			
P0001	S0102	2017-02-01	0	0.00	7	6.25	PR14		PR03			
P0001	S0103	2017-02-01	1	5.30	10	6.25	PR14		PR03			
P0001	S0104	2017-02-01	0	0.00	22	6.25	PR14		PR03			
P0001	S0105	2017-02-01	0	0.00	6	6.25	PR14		PR03			
P0001	S0106	2017-02-01	1	5.30	3	6.25	PR14		PR03			
P0001	S0107	2017-02-01	0	0.00	2	6.25	PR14		PR03			
P0001	S0108	2017-02-01	0	0.00	7	6.25	PR14		PR03			
P0001	S0110	2017-02-01	0	0.00	1	6.25	PR14		PR03			
P0001	S0115	2017-02-01	0	0.00	1	6.25	PR14		PR03			
P0001	S0122	2017-02-01	0	0.00	1	6.25	PR14		PR03			
P0001	S0125	2017-02-01	0	0.00	7	6.25	PR14		PR03			
P0001	S0131	2017-02-01	0	0.00	2	6.25	PR14		PR03			
P0001	S0132	2017-02-01	0	0.00	5	6.25	PR14		PR03			
P0001	S0142	2017-02-01	0	0.00	1	6.25	PR14		PR03			
P0004	S0006	2017-02-01	0	0.00	18	4.50	PR14		PR03			
P0004	S0019	2017-02-01	0	0.00	2	4.50	PR14		PR03			
P0004	S0038	2017-02-01	0	0.00	2	4.50	PR14		PR03			
P0004	S0044	2017-02-01	1	3.81	7	4.50	PR14		PR03			
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P0005	S0001	2017-02-01	0	0.00	7	33.90	PR14		PR03			
P0005	S0002	2017-02-01	0	0.00	11	33.90	PR14		PR03			
P0005	S0008	2017-02-01	0	0.00	3	33.90	PR14		PR03			
P0005	S0012	2017-02-01	0	0.00	6	33.90	PR14		PR03			
P0005	S0013	2017-02-01	0	0.00	11	33.90	PR14		PR03			
P0005	S0020	2017-02-01	0	0.00	3	33.90	PR14		PR03			
P0005	S0023	2017-02-01	0	0.00	3	33.90	PR14		PR03			
P0005	S0024	2017-02-01	0	0.00	3	33.90	PR14		PR03			

sales | sales_pivot | product_hierarchy | product_names | store_cities | store_names | city_names

Ready | Accessibility: Investigate | Count: 50000

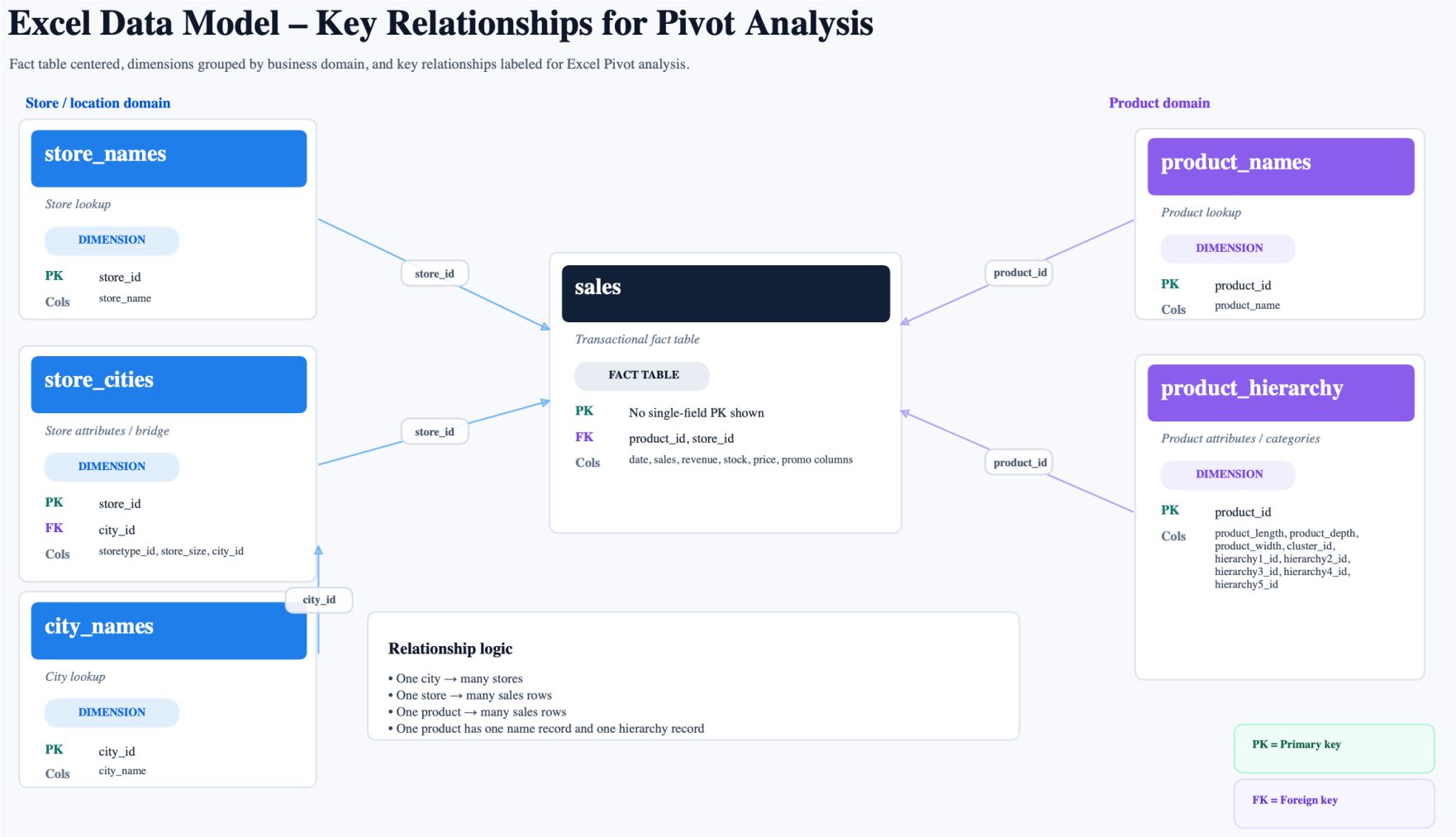
Data after cleaning – 49290 records

product_id	store_id	date	sales	revenue	stock	price	promo_type_1	promo_bin_1	promo_type_2
P0005	S0001	2017-02-01	0	0.00	7	33.90	PR14		PR03
P0005	S0001	2017-03-01	0	0.00	7	33.90	PR14		PR03
P0005	S0001	2017-04-01	0	0.00	7	33.90	PR14		PR03
P0005	S0001	2017-05-01	0	0.00	7	33.90	PR14		PR03
P0011	S0001	2017-02-01	0	0.00	10	49.90	PR14		PR03
P0011	S0001	2017-03-01	0	0.00	10	49.90	PR14		PR03
P0011	S0001	2017-04-01	0	0.00	10	49.90	PR14		PR03
P0011	S0001	2017-05-01	0	0.00	10	49.90	PR14		PR03
P0015	S0001	2017-05-01	0	0.00	21	2.60	PR10	low	PR03
P0015	S0001	2017-02-01	1	2.41	20	2.60	PR14		PR03
P0015	S0001	2017-03-01	0	0.00	20	2.60	PR14		PR03
P0015	S0001	2017-04-01	0	0.00	21	2.60	PR14		PR03
P0017	S0001	2017-05-01	0	0.00	12	1.49	PR06	high	PR03
P0017	S0001	2017-02-01	0	0.00	13	1.49	PR14		PR03
P0017	S0001	2017-03-01	0	0.00	13	1.49	PR14		PR03
P0017	S0001	2017-04-01	1	1.38	12	1.49	PR14		PR03
P0018	S0001	2017-02-01	0	0.00	49	1.95	PR14		PR03
P0018	S0001	2017-03-01	0	0.00	49	1.95	PR14		PR03
P0018	S0001	2017-04-01	0	0.00	57	1.95	PR14		PR03
P0018	S0001	2017-05-01	0	0.00	57	1.95	PR14		PR03
P0024	S0001	2017-02-01	0	0.00	23	1.95	PR14		PR03
P0024	S0001	2017-03-01	0	0.00	23	1.95	PR14		PR03
P0024	S0001	2017-04-01	0	0.00	23	1.95	PR14		PR03
P0024	S0001	2017-05-01	0	0.00	23	1.95	PR14		PR03
P0035	S0001	2017-02-01	2	4.54	14	2.45	PR14		PR03
P0035	S0001	2017-03-01	0	0.00	14	2.45	PR14		PR03
P0035	S0001	2017-04-01	0	0.00	14	2.45	PR14		PR03
P0035	S0001	2017-05-01	2	4.54	12	2.45	PR14		PR03
P0046	S0001	2017-02-01	0	0.00	4	34.50	PR14		PR03
P0046	S0001	2017-03-01	0	0.00	4	34.50	PR14		PR03
P0046	S0001	2017-04-01	0	0.00	4	34.50	PR14		PR03
P0046	S0001	2017-05-01	0	0.00	4	34.50	PR14		PR03
P0051	S0001	2017-02-01	0	0.00	27	0.70	PR14		PR03
P0051	S0001	2017-03-01	3	1.94	24	0.70	PR14		PR03
P0051	S0001	2017-04-01	0	0.00	24	0.70	PR14		PR03
P0051	S0001	2017-05-01	7	4.54	17	0.70	PR14		PR03
P0054	S0001	2017-02-01	0	0.00	6	3.95	PR14		PR03
P0054	S0001	2017-03-01	0	0.00	6	3.95	PR14		PR03
P0054	S0001	2017-04-01	0	0.00	6	3.95	PR14		PR03
P0054	S0001	2017-05-01	0	0.00	6	3.95	PR14		PR03
P0055	S0001	2017-02-01	0	0.00	14	3.50	PR05	verylow	PR03
P0055	S0001	2017-03-01	0	0.00	14	3.50	PR05	verylow	PR03
P0055	S0001	2017-04-01	4	10.37	10	3.50	PR05	verylow	PR03
P0055	S0001	2017-05-01	0	0.00	10	3.50	PR14		PR03
P0057	S0001	2017-02-01	0	0.00	6	12.90	PR14		PR03
P0057	S0001	2017-03-01	0	0.00	6	12.90	PR14		PR03
P0057	S0001	2017-04-01	0	0.00	6	12.90	PR14		PR03
P0057	S0001	2017-05-01	0	0.00	6	12.90	PR14		PR03

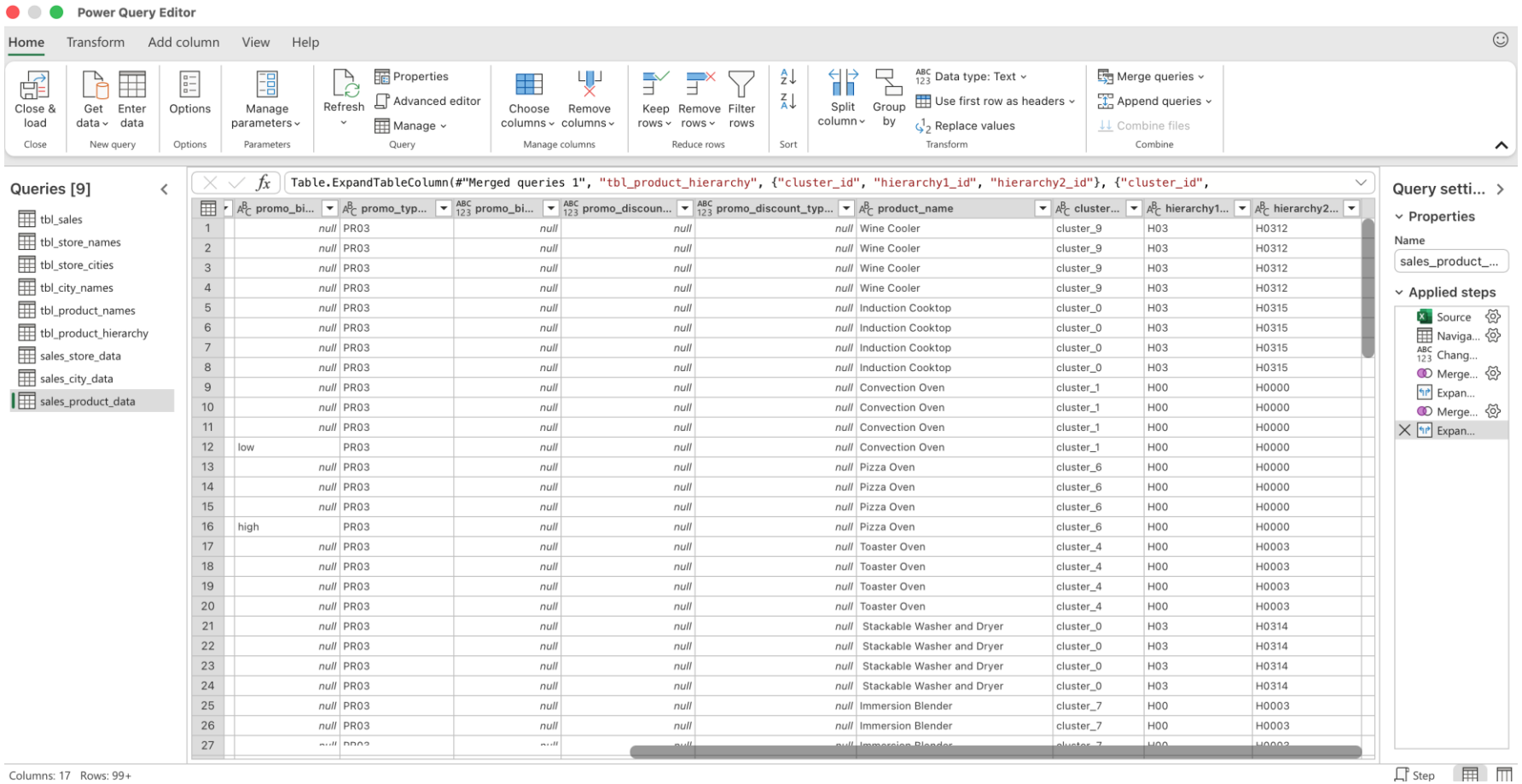
sales | product_hierarchy | product_names | store_cities | store_names | city_names

Ready | Accessibility: Investigate | Count: 49290

Appendix 2


Excel Data Model – Key Relationships for Pivot Analysis
Fact table centered, dimensions grouped by business domain, and key relationships labeled for Excel Pivot analysis.
Store / location domain
store_names
Store lookup
DIMENSION
PK store_id
Cols store_name
store_cities
Store attributes / bridge
DIMENSION
PK store_id
FK city_id
Cols storetype_id, store_size, city_id
city_names
City lookup
DIMENSION
PK city_id
Cols city_name
store_id
store_id
city_id
sales
Transactional fact table
FACT TABLE
PK No single-field PK shown
FK product_id, store_id
Cols date, sales, revenue, stock, price, promo columns
product_id
product_id
Product domain
product_names
Product lookup
DIMENSION
PK product_id
Cols product_name
product_hierarchy
Product attributes / categories
DIMENSION
PK product_id
Cols product_length, product_depth, product_width, cluster_id, hierarchy1_id, hierarchy2_id, hierarchy3_id, hierarchy4_id, hierarchy5_id
Relationship logic
• One city → many stores
• One store → many sales rows
• One product → many sales rows
• One product has one name record and one hierarchy record
PK = Primary key
FK = Foreign key

Appendix 3



	promo_bi...	promo_typ...	promo_bi...	promo_discoun...	promo_discount_typ...	product_name	cluster...	hierarchy1...	hierarchy2...
1	null	PR03	null	null	null	Wine Cooler	cluster_9	H03	H0312
2	null	PR03	null	null	null	Wine Cooler	cluster_9	H03	H0312
3	null	PR03	null	null	null	Wine Cooler	cluster_9	H03	H0312
4	null	PR03	null	null	null	Wine Cooler	cluster_9	H03	H0312
5	null	PR03	null	null	null	Induction Cooktop	cluster_0	H03	H0315
6	null	PR03	null	null	null	Induction Cooktop	cluster_0	H03	H0315
7	null	PR03	null	null	null	Induction Cooktop	cluster_0	H03	H0315
8	null	PR03	null	null	null	Induction Cooktop	cluster_0	H03	H0315
9	null	PR03	null	null	null	Convection Oven	cluster_1	H00	H0000
10	null	PR03	null	null	null	Convection Oven	cluster_1	H00	H0000
11	null	PR03	null	null	null	Convection Oven	cluster_1	H00	H0000
12	low	PR03	null	null	null	Convection Oven	cluster_1	H00	H0000
13	null	PR03	null	null	null	Pizza Oven	cluster_6	H00	H0000
14	null	PR03	null	null	null	Pizza Oven	cluster_6	H00	H0000
15	null	PR03	null	null	null	Pizza Oven	cluster_6	H00	H0000
16	high	PR03	null	null	null	Pizza Oven	cluster_6	H00	H0000
17	null	PR03	null	null	null	Toaster Oven	cluster_4	H00	H0003
18	null	PR03	null	null	null	Toaster Oven	cluster_4	H00	H0003
19	null	PR03	null	null	null	Toaster Oven	cluster_4	H00	H0003
20	null	PR03	null	null	null	Toaster Oven	cluster_4	H00	H0003
21	null	PR03	null	null	null	Stackable Washer and Dryer	cluster_0	H03	H0314
22	null	PR03	null	null	null	Stackable Washer and Dryer	cluster_0	H03	H0314
23	null	PR03	null	null	null	Stackable Washer and Dryer	cluster_0	H03	H0314
24	null	PR03	null	null	null	Stackable Washer and Dryer	cluster_0	H03	H0314
25	null	PR03	null	null	null	Immersion Blender	cluster_7	H00	H0003
26	null	PR03	null	null	null	Immersion Blender	cluster_7	H00	H0003
27	null	PR03	null	null	null	Immersion Blender	cluster_7	H00	H0003

Power queries:

Merging queries to facilitate enhancing data analysis using Microsoft Excel

Appendix 4

Query Query History

```
SELECT * FROM product_hierarchy LIMIT 5;
```

Data Output Messages Notifications

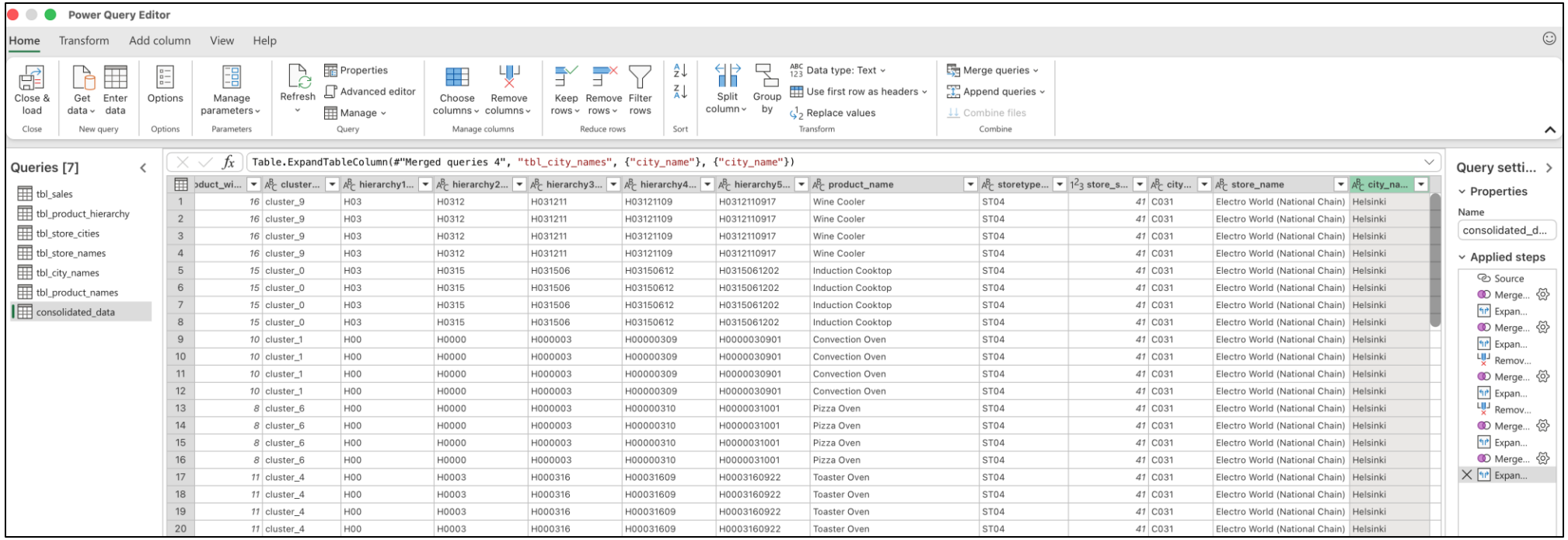
	product_id character varying	product_length double precision	product_depth double precision	product_width double precision	cluster_id character varying
1	P0000	5	20	12	[null]
2	P0001	13.5	22	20	cluster_5
3	P0002	22	40	22	cluster_0
4	P0004	2	13	4	cluster_3
5	P0005	16	30	16	cluster_9

Total rows: 5 of 5 Query complete 00:00:00.228 Ln 1, Col 32

SQL

Load first 5 product_hierarchy rows

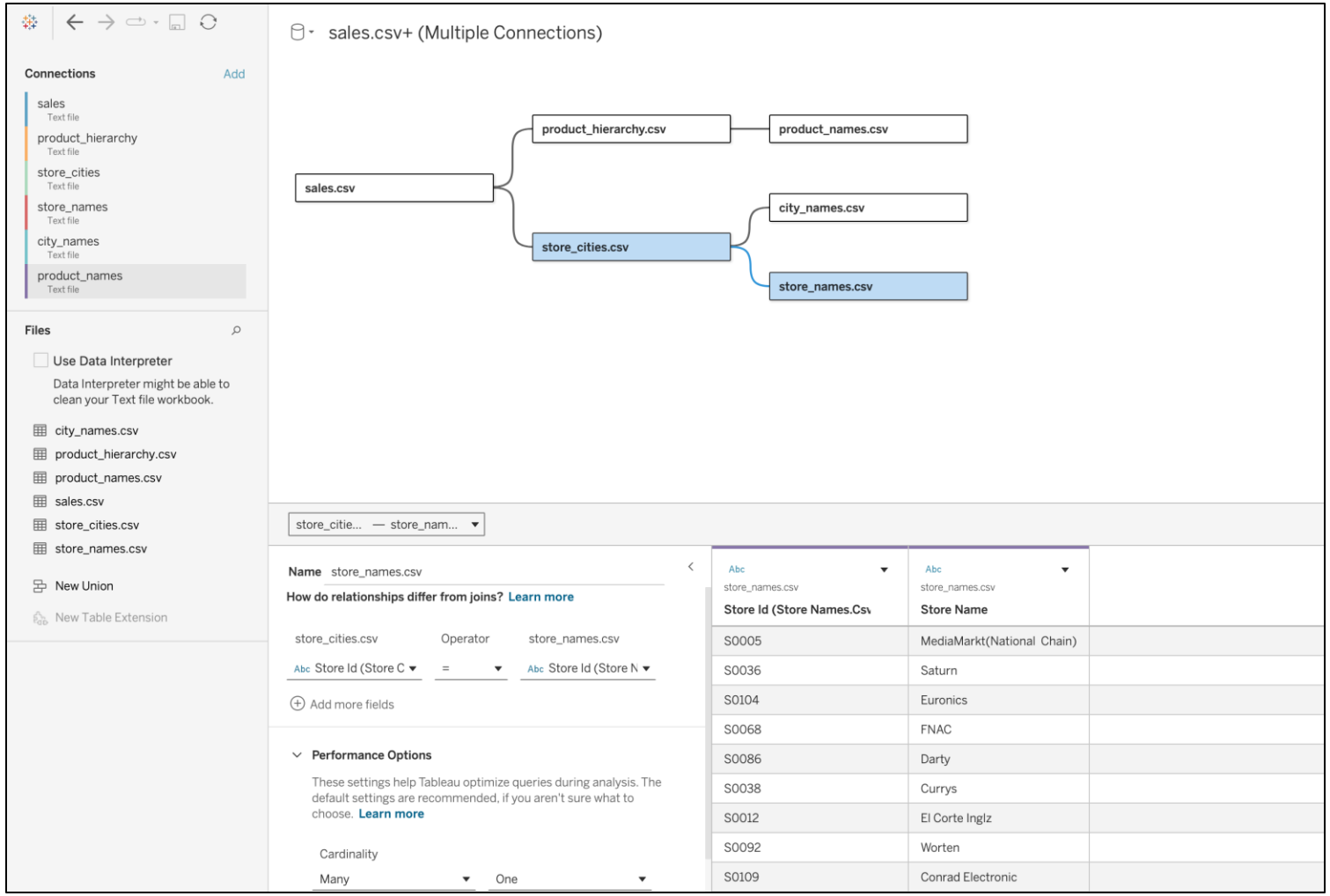
Appendix 5



Power queries:

Merging queries to create consolidated data worksheet

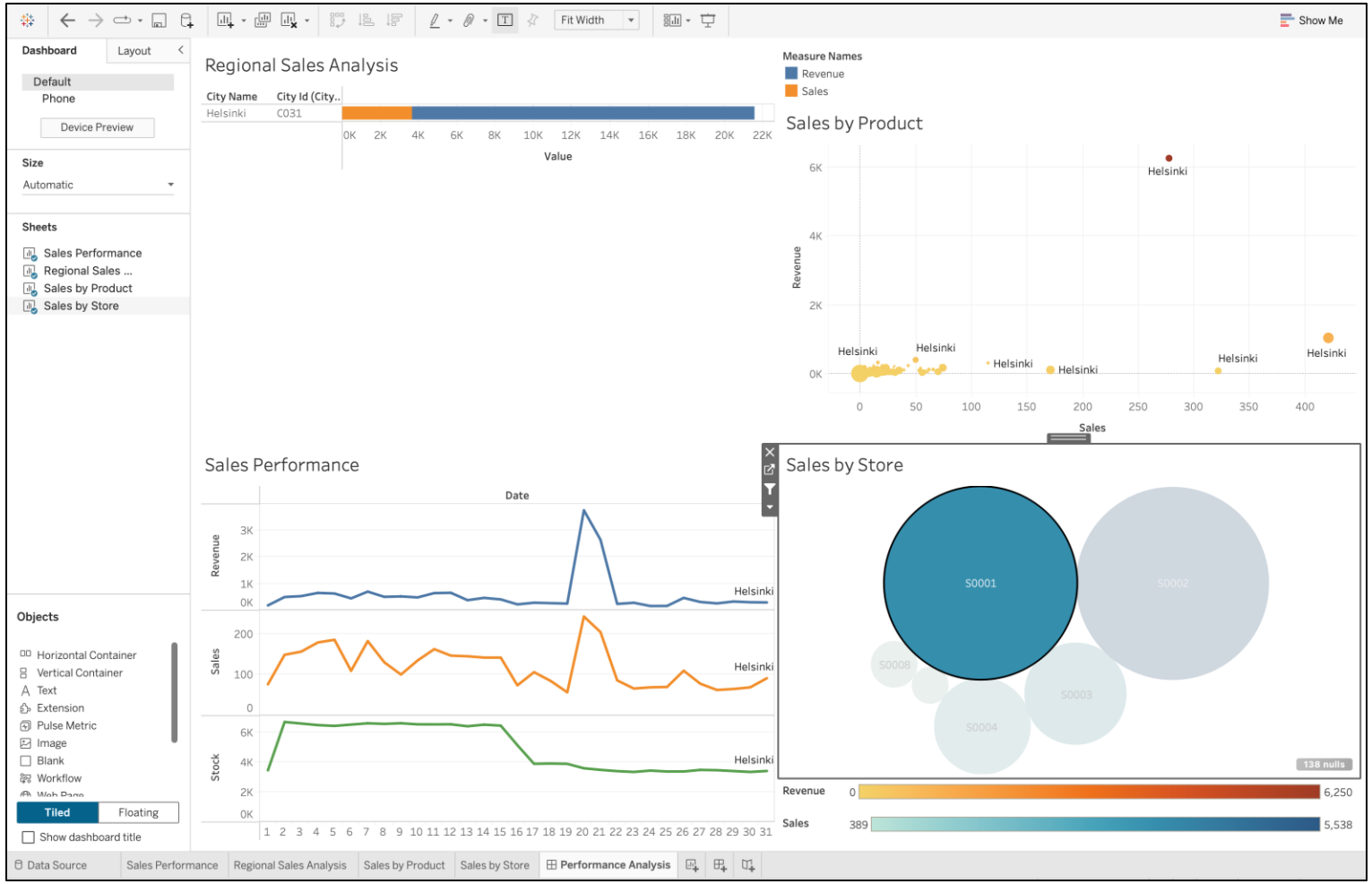
Appendix 6



Tableau

Relationships and cardinality

Appendix 7



Tableau

Filtered dashboard